Neutrino Oscillation

- Assume two massive neutrino states ν_1, ν_2. Then,

$$|\nu_e\rangle = \cos\theta \cdot |\nu_1\rangle - \sin\theta \cdot |\nu_2\rangle$$

$$|\nu_\mu\rangle = \sin\theta \cdot |\nu_1\rangle + \cos\theta \cdot |\nu_2\rangle$$

- The lepton mixing matrix

$$V = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Neutrino Oscillation

In vacuum:

$$P_{ee} = 1 - \sin^2 2\theta \sin^2 \frac{\Delta m^2 L}{4E}$$

$$P_{e\mu} = \sin^2 2\theta \sin^2 \frac{\Delta m^2 L}{4E}$$

Practically useful:

$$P_{e\mu} = \sin^2 2\theta \;\; \sin^2\left(1.27 \, \frac{\Delta m^2}{10^{-3}\,\mathrm{eV}^2} \cdot \frac{L}{\mathrm{km}} \cdot \frac{\mathrm{MeV}}{E} \right)$$